UniHyper: A Unified Multimodal Classification Framework for Hyperspectral Images Under Diff-Labeled Scenes

Xiaoyu Tian, *Student Member, IEEE*, Yi Liu, *Student Member, IEEE*, Chuan Fu, *Member, IEEE*, Yule Duan, *Member, IEEE*, Tan Guo, *Senior Member, IEEE*, and Fulin Luo, *Senior Member, IEEE*

***Abstract*—Land-cover classification using multimodal remote sensing imagery is a fundamental task in Earth observation. Yet, current methods face two major challenges: the tension between modality consistency and specificity preservation in feature fusion, and the lack of a universal framework adaptable to diverse labeled scenarios (e.g., fully supervised, semi-supervised, noisy-label, weakly supervised, and unsupervised settings). To address these issues, this article proposes UniHyper, a unified multimodal classification framework that explicitly separates representation learning from the subsequent classification process. For representation learning, we introduce a Fourier-orthogonal feature decoupling strategy (FoFD) strategy to separate modality-invariant and modality-unique features, mitigating the specificity erosion problem in contrastive learning. Furthermore, a Fourier-orthogonal channel attention network (FoCA-Net) is designed to enhance cross-modal consistency while preserving discriminative characteristics via frequency-domain attention. For classification, a flexible teacher–student classifier is developed to process the different supervised tasks, while a dynamic prototype clustering strategy is proposed for the unsupervised task. Extensive experiments on multimodal remote sensing datasets demonstrate that UniHyper achieves state-of-the-art performance across multiple labeled scenarios, offering a generalized solution for robust land-cover classification.**

***Index Terms*—Feature decoupling, land-cover classification, multimodal remote sensing, teacher–student learning, unified framework, unsupervised clustering.**

I. Introduction

LAND-COVER classification is one of the earliest and most widely used technologies in remote sensing, primarily conducted using hyperspectral image (HSI) [1], [2], [3], [4]. It aims to categorize each pixel in HSI into its corresponding land-cover class based on its spectral characteristics [5]. With the rapid advancement of aerospace satellites and spacecraft, the technology for synchronously acquiring multimodal remote sensing imagery from multiple platforms within the same scene is increasingly feasible. Multimodal remote sensing imagery captures distinct attributes of the land cover [6], [7]. For example, the HSI, comprising hundreds of spectral bands, provides detailed spectral information about the land cover [8], [9]. The synthetic aperture radar (SAR) captures structural features under all weather conditions, enabling continuous global observation [10], [11]. The light detection and ranging (LiDAR) captures precise 3-D information, allowing detailed characterization of target features [12], [13]. Similarly, different modalities also exhibit distinct limitations. HSI is susceptible to weather conditions and struggles to distinguish objects with similar spectral reflectance. SAR faces challenges in differentiating intricate structures. LiDAR suffers from weak object classification capabilities. Evidently, images acquired from different modalities are both enabled and constrained by their distinct imaging mechanisms, thereby providing diverse and complementary information for Earth observation [14], [15], [16].

In recent years, leveraging the powerful feature extraction capabilities of deep neural networks, extensive research has been conducted on multimodal feature extraction for more precise land-cover identification [17], [18], [19], [20], [21], [22]. The most common multimodal classification algorithms currently fuse information extracted from various modalities to improve object distinction [23]. For example, FusAtNet [24] employs a feature fusion and extraction framework for collective land-cover classification of HSIs and LiDAR data. S3F2Net [15] designs a spatial–spectral–structural feature fusion network to extract rich features of multimodal data from different perspectives. However, such methods neglect the correlations between modalities, thereby limiting the model's classification performance [25]. To address this, researchers employ contrastive learning frameworks to enhance the feature consistency across multimodal data [26], [27]. MCFT [28] proposed a multimodal contrastive fusion Transformer for effective multimodal data fusion. CAMFNet [29] introduces a novel semantic similarity contrastive loss that fully utilizes a large amount of unlabeled data to learn potential complementary information between different modalities. However, these methods suffer from specificity erosion, where an excessive pursuit of feature consistency across modalities suppresses the unique characteristics of each modality, leading to overly

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Xiaoyu Tian, Yi Liu, Chuan Fu, and Fulin Luo are with the College of Computer Science, Chongqing University, Chongqing 400044, China (e-mail: xiaoyu96@stu.cqu.edu.cn; liuyi@stu.cqu.edu.cn; fuchuan@cqu.edu.cn; luoflyn@163.com).

Yule Duan is with the College of Information Science and Engineering, Henan University of Technology, Zhengzhou 450001, China (e-mail: duanyule@haut.edu.cn).

Tan Guo is with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China, and also with the Institute of Industrial Internet, Chongqing University of Posts and Telecommunications, Chongqing 401122, China (e-mail: guot@cqupt.edu.cn).

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homogenized representations [30], [31]. For instance, in the fusion of SAR and HSI data, the geometric cues provided by SAR may be overwhelmed by the rich spectral features of HSI, resulting in the loss of complementary modality-unique information. This phenomenon reveals a fundamental tension in existing multimodal learning approaches: an imbalance between optimizing modality-consistent features and preserving modality-unique features.

Furthermore, compared to other computer vision tasks, land-cover classification requires labeling every pixel in an image [32]. This process is not only time-consuming and labor-intensive but also requires a higher level demand of specialized knowledge [33], [34]. To address the challenges posed by labeling difficulties in terms of both quality and quantity, researchers have developed several learning paradigms—including semi-supervised learning [35], [36], [37], [38], noisy labeling learning [39], [40], [41], [42], weakly supervised learning [43], [44], [45], and unsupervised learning [46], [47]—which have mitigated the constraints imposed by data limitations on feature classification performance to a certain extent. However, the above methods are all designed for a single-labeled scenario and cannot be successfully transferred to other labeled scenarios, failing to form a universal solution. For instance, existing semi-supervised learning models can only leverage large amounts of unlabeled samples to improve classification performance when labeled data are scarce and cannot achieve satisfactory performance in scenarios involving noisy or weak annotations. There is an urgent issue that current research lacks generalizability and applicability across different labeled scenarios. In various labeled scenarios, many land-cover classification algorithms remain tailored to single-modal HSIs. This reliance on unimodal features restricts representation diversity and further undermines model generalization under complex annotation conditions.

Based on the above analysis, it is necessary to integrate multimodal data to enhance the feature representations and contribute to the model's generalizability under diff-labeled scenes. Thus, we propose a unified multimodal classification framework applicable to diverse labeled scenarios. Within this framework, we decouple the representation process from the classification task for the first time. Specifically, during the multimodal representation process, we explicitly decouple features across modalities into modality-common features and modality-unique features. Based on contrastive learning theory, we constrain the consistency among modality-common features and the complementarity among modality-unique features, maintaining a balance between shared and specific features to extract effective information across modalities fully. During the classification process, we designed a teacher–student classifier independent of the representation process based on the teacher–student learning framework. This enables the student classifier to obtain effective supervisory information from the teacher classifier under any annotation scenario, achieving precise feature classification. Additionally, to address unsupervised clustering tasks, we developed a dynamic prototype clustering technique that enables large-scale unsupervised classification. Our key contributions are summarized as follows.

1) We propose a unified multimodal classification framework for land-cover classification, UniHyper, which explicitly decouples the representation learning process and classification task and can be applied to diverse labeled scenes.
2) We design a Fourier-orthogonal channel attention network (FoCA-Net) that leverages random Fourier-orthogonal filters (OFs) for frequency-domain attention, enhancing multimodal feature extraction to obtain discriminative characteristics.
3) We design a Fourier-orthogonal feature decoupling strategy (FoFD), which decouples modality-common and modality-unique features to mitigate the specificity erosion problem inherent in contrastive learning networks.
4) We introduce the teacher–student classifier for land-cover classification under different supervision information and the dynamic prototype clustering strategy for unsupervised land-cover classification.

The organization of the rest of this article is as follows. Section II details the specific framework. Section III conducts some experiments under different labeled scenarios to demonstrate the effectiveness of the proposed framework. In Section IV, we analyze some conclusions of this article and suggestions for future work.

II. Method

Multimodal classification in remote sensing refers to extracting feature representations from different image types and identifying the land-cover type of each pixel in an image. Let $D = \{(x_i, y_i)\}_{i=1}^{N}$ denotes the multimodal datasets, and each pair $x_i = (x_i^1, \ldots, x_i^m, x_i^s)$ consists $m + 1$ madalities and a corresponding label $y_i \in Y$, where Y refers the the label space with M labels. Specifically, $\{x_i^v\}_{v=1}^{m}$ represents data patches extracted from the multimodal remote sensing image with a sliding window; x_i^s denotes the spectral information from HSI. Our goal is to train a unified multimodal classification framework that can achieve satisfactory performance under diff-labeled scenarios via multimodal representation learning.

The framework of UniHyper is as in Fig. 1, which consists of three specific processes: multimodal feature representation, land-cover identification, and diff-labeled optimization. Since the focus of this article is the model's versatility, we employ a simple workflow for land-cover classification under diff-labeled scenes. At the same time, we emphasize research and propose universally adaptable feature representation and classification mechanisms. The details of implementation are as follows.

A. Multimodal Feature Representation

In current studies, multimodal representation learning methods have overly pursued intermodal feature consistency, leading to specificity erosion. Furthermore, current multimodal feature extraction techniques mainly suffer from the high computational complexity and insufficient cross-modal fusion

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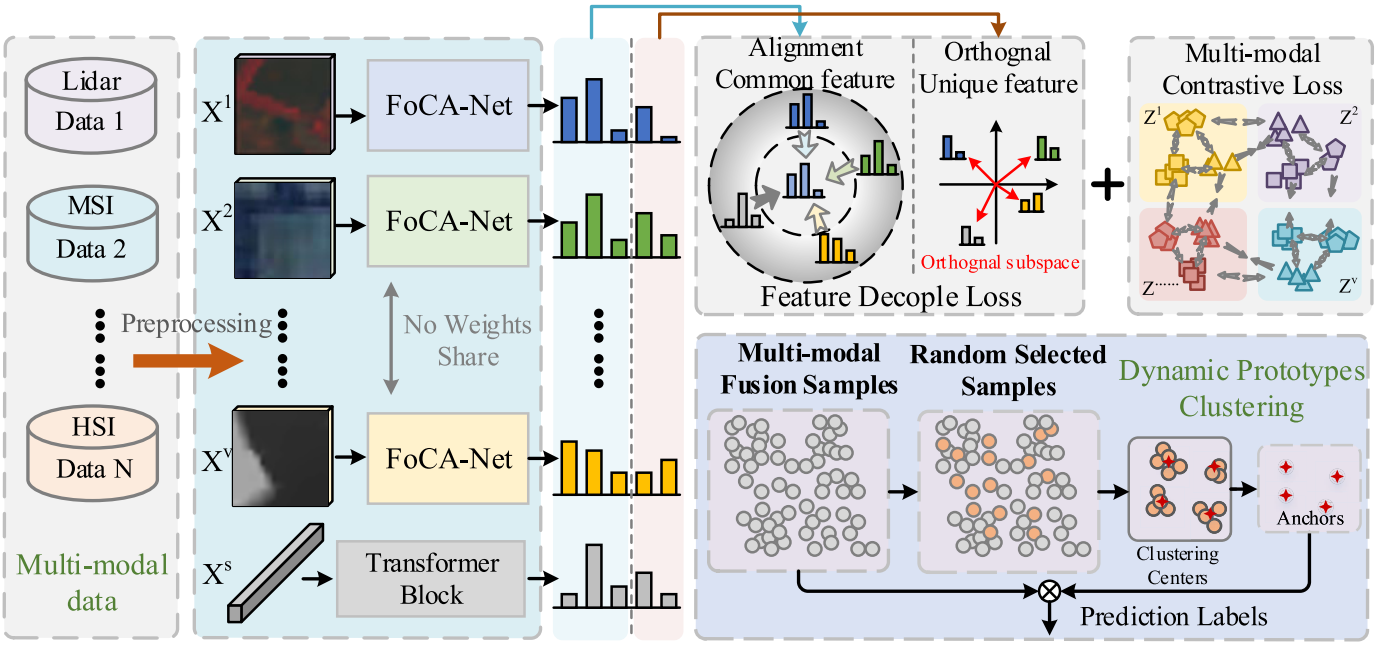


Fig. 1. Illustration of UniHyper architecture.

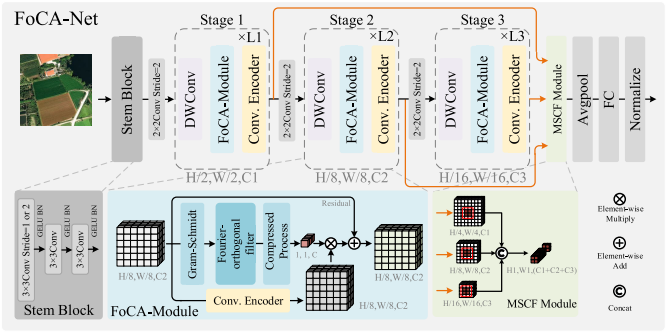


Fig. 2. Illustration of FoCA-Net architecture.

capability of Transformer, which motivates us to build an efficient multistage FoCA-Net-based CMT structure [48] for efficient multimodal feature extraction.

Fourier-Orthogonal Channel Attention Network: As shown in Fig. 2, we first obtain the coarse features $\hat{x}_i^v$ for vth modalities of the ith sample x_i^v via the stem block, which consists of three convolution layers. To extract richer features, we propose a novel Fourier-orthogonal channel attention mechanism (FoCA) based on OrthoNets [49], encapsulating it into three stages with convolutional networks. In FoCA, we leverage the random OFs to obtain the spatial information of each feature, as shown in the following:

$$\mathbf{F}_o = \phi_{GS}(\mathbf{R}) \odot \hat{x}_i^v \tag{1}$$

where ϕ_{GS} denotes the Gram–Schmidt process and $\mathbf{R}$ is the initialized random filter bank. Then, we transform $\mathbf{F}_o$ into the frequency domain and retain a low-frequency region controlled by the masking ratio η. Specifically, let Ω_η denote the centered low-frequency region whose area ratio is η. The binary low-frequency mask is defined as

$$M_{low}(u,v) = \mathbb{I}\left[(u,v) \in \Omega_\eta\right] \tag{2}$$

where (u, v) denotes a frequency coordinate and η controls the proportion of retained low-frequency components. The Fourier-OF $\mathbf{F}_f$ is then calculated by

$$\mathbf{F}_f = \mathcal{F}^{-1}\left(M_{low} \odot \mathcal{F}(\mathbf{F}_o)\right). \tag{3}$$

Then, the Fourier-OF $\mathbf{F}_f$ is compressed to a channel vector to yield the attention weight $\mathcal{A}$ as shown in the following:

$$\mathcal{A} = \sigma\left(W_2\delta\left(W_1\mathbf{F}_f\right)\right) \tag{4}$$

where $\sigma = Sigmoid(\cdot)$, δ represents the ReLU activation function, and W_1 and W_2 are weight matrices. The output features are obtained by scaling the input features x_i with the attention vector $\mathcal{A}$ and adding a residual connection to facilitate gradient flow, as shown in the following:

$$Z_o = \text{Conv}\left(\hat{x}_i^v\right) \odot \mathcal{A} + \hat{x}_i^v. \tag{5}$$

Multiscale Central-Pixel Fusion (MSCF) Module: Given the data patch x_i^v for the vth modality of the ith sample, including the center pixel and its neighbor pixels, we assume that the characteristics of the central region can more accurately describe the categorical features of the sample. Thus, after gradually extracting deep semantic features $\hat{x}_{ij}^v = Stage_j(\hat{x}_{i(j-1)}^v) \in \mathbb{R}^{C_j \times H_j \times W_j}$ through network modules at different stages (such as convolution, attention mechanism, etc.), we preserve the most representative feature by cropping the central area in the MSCF module, as shown in the following:

$$\hat{x}_j^{crop} = \hat{x}_{ij}^v\left[:,:,\left\lfloor\frac{H_i}{2}\right\rfloor - 1 : \left\lfloor\frac{H_i}{2}\right\rfloor, \left\lfloor\frac{W_i}{2}\right\rfloor - 1 : \left\lfloor\frac{W_i}{2}\right\rfloor\right] \tag{6}$$

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where $\lfloor\cdot\rfloor$ represents the flooring function. Then, we concatenate the outputs of each stage in the channel dimension as shown in the following:

$$x_{\text{fused}} = \oplus_{i=1}^{n}\left(\hat{x}_j^{crop}\right) \in \mathbb{R}^{B\times\left\{\sum_{i=1}^{n} D_i\right\}} \tag{7}$$

where $\oplus$ indicates the tensor concat operation, B is the batch size, and D_i is the number of the output features at the ith stage. Finally, FFN is used to obtain the latent features of each modality: $z_i^v = \text{FFN}(x_{\text{fused}}) \in R^d$, where d is the feature dimension. In the mini-bath with B samples, the samples' features of vth modality is $Z^v = \{z_1^v, \ldots, z_B^v\} \in R^{B\times d}$. For the spectral modality x_i^s, we directly capture its global related information Z^s via Transformer.

Multimodal Feature Decouple Loss: Based on the previous analysis, alleviating the issue of modality-unique erosion is crucial for achieving better multimodal representation. Given the representations $Z^v \in R^{B\times d}$ of modality v, we assume that the feature dimension of modality-common feature is $d_c < d$. Then, we split the first d_c-dimensional features $Z_{:d_c}^v \in R^{B\times d_c}$ as the modality-common feature and the remaining dimensions $Z_{d_c:}^v \in R^{B\times d-d_c}$ as the modality-unique feature. Based on this, we construct a feature decoupling loss to establish consistency constraints for common features between diverse modalities while preserving unique features specific to each modality. The principle of the feature decoupling loss is to draw closer the common feature among modalities and push apart the unique feature among modalities, which is as follows:

$$\mathcal{L}_{FD} = \lambda_1\mathcal{L}_{cf} + \lambda_2\mathcal{L}_{uf} \tag{8}$$

where λ_1 and λ_2 are balance coefficients, $\mathcal{L}_c$ is the common feature loss, and $\mathcal{L}_p$ is the unique feature loss.

1) *Common Feature Alignment Loss:* Ideally, the common feature representations among diverse modalities should be consistent and drawn closer, that is, the similarity between positive sample pairs (different modalities of the same instance) should be close to 1, while the similarity between negative sample pairs (arbitrary modalities of different instances) should be close to 0. Therefore, we expect that the similarity matrix $C_{ij} = (\langle Z_{:d_c}^i, Z_{:d_c}^j\rangle)/(\|Z_{:d_c}^i\|\|Z_{:d_c}^j\|)$ should be close to the identity matrix I, and the common feature alignment loss function is as follows:

$$\mathcal{L}_{cf} = \sum_{i=1}^{v}\sum_{j\neq i}\|C_{ij} - I\|_F^2. \tag{9}$$

2) *Unique Feature Orthogonal Loss:* We encourage the unique feature representation of each modality for the same instance to be different from each other. In other words, we hope that the distance between the unique features of diverse modalities would be large enough, that is, to pull away the unique feature representation of each modality. To this end, we directly calculate the Euclidean distance of the unique feature $u_{ij} = \|Z_{d_c:}^{i\top} - Z_{d_c:}^{j\top}\|_2$, and the unique feature loss function is as follows:

$$\mathcal{L}_{uf} = \sum_{i=1}^{v}\sum_{j\neq i}\left(1 - \sigma\left(u_{ij}\right)\right) \tag{10}$$

where $\sigma(x) = 1/(1+e^{-x})$ represents the sigmoid function and $\mathcal{L}_p$ maps the Euclidean distance to $(0, 1)$. When $u \to +\infty$, $\sigma(u) \to 1$, the loss approaches 0. Ideally, the unique feature representation of each modality of the same instance should satisfy $Z_{d_c:}^i \perp Z_{d_c:}^j$, $\|Z_{d_c:}^j| \gg 0$ in the subspace.

B. Land-Cover Identification

To overcome the limitation of existing classification methods, which are only adaptive for one annotation scene, we design a teacher–student classifier and dynamic prototype clustering for land-cover identification with and without supervision, respectively.

Teacher–Student Classifier: As shown in Fig. 3(a), we first concatenate the decoupled multimodal features to preserve the unique features of each modality as much as possible

$$Z = \left[\frac{1}{m+1}\sum_{i=1}^{m+1}\underbrace{Z_{:d_c}^i}_{d_c} \| \oplus_i^{m+1}\left(\underbrace{Z_{d_c:}^i}_{(d-d_c)}\right)\right] \tag{11}$$

where $\oplus$ denotes channel-wise concatenation and Z is the feature representation for classification in the mini-batch with the sample size of B. Then, we directly put Z into the teacher classifier with the parameters θ_t and add the random noise to Z for the student classifier with the parameters θ_s, which can be expressed as

$$\begin{cases} p^t = Softmax(W_t Z) \\ p^s = Softmax(W_s(Z + w\alpha)) \end{cases} \tag{12}$$

where α is the random noise; w is the weight value between 0 and 1; and W_t and W_s denote the parameters of the teacher classifier and student classifier, respectively. The parameters W_s of the student classifier are updated via standard gradient descent. After each training step, the parameters W_t of the teacher classifier are updated using an exponential moving average of the student parameters. To enhance the predicted accuracy, we measure the consistency between the teacher classifier and the student classifiers via Kullback–Leibler divergence as follows:

$$\mathcal{L}_{kl} = \mathcal{D}_{KL}\left(p^s \| p^t\right). \tag{13}$$

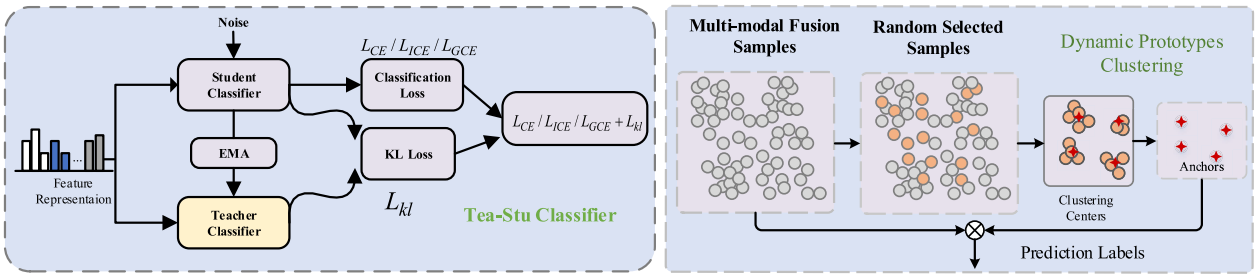
Dynamic Prototype Clustering: After obtaining highly discriminative multimodal feature representations in a unified latent space via (11), dynamic prototype clustering method is used to cluster the randomly sampled samples $\mathcal{B} = \{Z_i | i \sim Uniform(1, N)\} \in \mathbb{R}^{\rho N\times(d_c+(m+1)\times(d-d_c))}$ to obtain the cluster centers

$$\mu = \text{K-means}\left(\mathcal{B}\right) \in \mathbb{R}^{K\times(d_c+(m+1)\times(d-d_c))}. \tag{14}$$

In this way, we can reduce the time complexity from (Nk^2) to (ρNk^2). Experimental verification shows that when $\rho = 0.1$, the multimodal clustering performance of our network does not decrease significantly, which demonstrates the clustering stability of our method. Then, to ensure strict one-to-one correspondence between cluster centroids and semantic prototypes and prevent semantic discontinuities caused by index misalignment, we employ the Hungarian algorithm [50] for optimal bipartite matching during prototype assignment. Then, we update the prototypes through momentum

$$\mathcal{C}^t = \alpha\mathcal{C}^{(t-1)} + (1-\alpha)\mu. \tag{15}$$

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(a) (b)

Fig. 3. Structure of land-cover identification. (a) Teacher–student classifier. (b) Dynamic prototype clustering.

Finally, we calculate the similarity between the feature and the prototype $\mathcal{C}$ and predict the most relevant label in the whole label space for the instance

$$y_i = \arg\max_j \left(Z_i (\mathcal{C})^T\right) \tag{16}$$

where Z_i is the feature representation for clustering of the ith sample in each mini-batch.

C. Diff-Labeled Optimization

Given the differences in supervision information across various annotation scenarios, we employ distinct losses under a unified representation and classification framework to optimize the network architecture. For the common feature representation learning under diff-labeled scenes, we optimize the multimodal representation framework proposed in this article through the contrastive loss, which is expressed as

$$\mathcal{L}_{cons} = \mathcal{L}_{MC} + \mathcal{L}_{FD} \tag{17}$$

where $\mathcal{L}_{FD}$ is the decouple loss in (10) and $\mathcal{L}_{MC}$ is the consistency loss based on SimCLR. Specifically, taking two modalities of the same sample to construct the positive pair (Z^i, Z^j), the goal is to maximize the similarities between positive pairs while minimizing the similarities of negative pairs. Then, the similarity between two features can be measured by the cosine distance as follows:

$$S\left(Z^i, Z^j\right) = \frac{\left\langle Z^i, Z^j \right\rangle}{\left\| Z^i \right\| \left\| Z^j \right\|} \tag{18}$$

where $\langle \cdot, \cdot \rangle$ represents the dot product operator. Then, the feature consistency objective $\mathcal{L}_{MC}$ can be formulated as

$$\mathcal{L}_{MC} = \sum_{i=1}^{m+1} \sum_{j \neq i} - \left[\log \frac{exp\left(S\left(Z^i, Z^j\right)\right)}{\sum_{(Z^i, Z^k) \in B} exp\left(S\left(Z^i, Z^k\right)\right)} \right] \tag{19}$$

where B denotes sample pairs in each batch.

Full-Supervised Learning: Fully supervised learning refers to the process of training a mapping function from input samples to output labels, where all training samples are precisely labeled. The optimization of the model parameters by minimizing the cross-entropy loss between the predicted probabilities and the ground truth labels across all training samples, as expressed by the following formula:

$$\mathcal{L}_{CE} = -\frac{1}{B} \sum_{i=1}^{B} \sum_{c=1}^{M} q_{ic} \cdot \log p_{ic} \tag{20}$$

where B is the batch size, M is the total number of categories, q_i is the one-hot encoded vector of y_i, q_{ic} is the one-hot value of sample i for category c, and p_{ic} is the probability predicted by the model that sample i belongs to category c.

We first obtain the classification resluts p^t and p^s via (12). Since all samples are accurately labeled, we directly compute cross-entropy loss $\mathcal{L}_{CE}$ between the student classifier's output p^s and the ground truth labels via (20). Additionally, we calculate consistency loss $\mathcal{L}_{kl}$ between the teacher and student classifiers' outputs p^t and p^s via (13) to enhance the classification performance. The total loss for full-supervised classification is the sum of $\mathcal{L}_{cons}$, $\mathcal{L}_{CE}$, and $\mathcal{L}_{kl}$, which can be expressed by

$$\mathcal{L}_{FS} = \mathcal{L}_{cons} + \mathcal{L}_{CE} + \mathcal{L}_{kl}. \tag{21}$$

Semi-Supervised Learning: Semi-supervised learning trains models using both labeled data and unlabeled data, typically implemented by the mean teacher model [51]. We first divide the sample into labeled data and unlabeled data. For labeled data, we compute the loss $\mathcal{L}_{FS}$ via (21) in the same way as full-supervised learning. For unlabeled data, based on the feature contrastive loss $\mathcal{L}_{cons}$, we leverage the teacher's output p^t to generate the pseudo labels $\hat{p}$ via FreeMatch method [52] and compute the cross-entropy loss between the student's output p^s and the pseudo labels $\hat{p}$. The specific formula is as follows:

$$\mathcal{L}_{uns} = \mathcal{L}_{cons} + \frac{1}{\hat{B}} \sum_{i=1}^{\hat{B}} \left(max\left(p_i^t\right) > \tau\right) \left(- \sum_{c=1}^{M} \hat{p}_{ic} \cdot \log p_{ic}^s \right) \tag{22}$$

where τ is the adaptive threshold the label space, refer to [52] for the specific process. $\hat{B}$ represents the number of samples in each batch for which the teacher classifier's predicted probability is greater than the threshold τ. The total loss of Semi-supervised learning loss is the sum of labeled data loss and unlabeled data loss, as shown in the following:

$$\mathcal{L}_{SL} = \mathcal{L}_{FS} + \mathcal{L}_{uns}. \tag{23}$$

Noisy-Supervised Learning: Noisy-supervised learning refers to situations where incorrect data annotation introduces

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label noise into training data—that is, incorrect label information. Given the noisy-labeled data $D = \{(x_i, \tilde{y}_i)\}_{i=1}^{N_l}$, where $\tilde{y}_i$ is the observed label. $\tilde{y}_i$ is the noisy label when $\tilde{y}_i$ is not equal to the ground truth label and vice versa. In the training process, the loss function should be robust to noise. The widely used loss for this task is the generalized cross-entropy loss as follows:

$$\mathcal{L}_{GCE} = \frac{1}{B}\sum_{i=1}^{B}\frac{1-\left(p_{i\tilde{y}}^{s}\right)^{\beta}}{\beta} \tag{24}$$

where β is a hyperparameter between 0 and 1. The closer the value of β is to 1, the higher its tolerance for label noise. Conversely, the stricter it is regarding label noise. In most cases, it is typically set to 0.7. To correct the noisy label, we also accumulate the momentum v_i of the teacher classifier's output p_i^t using the EMA method, with the detailed process given as follows:

$$v_i = \begin{cases} p_i^t, & if\ epoch = 1 \\ (1-\theta)\, v_i + \theta p_i^t, & otherwise. \end{cases} \tag{25}$$

Then, the label correction can be as follows:

$$\tilde{y}_i = \begin{cases} argmax\,(v_i), & if\ argmax\,(v_i) > \tau \\ \tilde{y}_i, & otherwise \end{cases} \tag{26}$$

where τ is the dynamic threshold for label correction; its value increases from 0.7 to 0.9 as the training batches grow.

For noisy label, we directly leverage the sum of the feature contrastive loss, the consistency loss in (13), and the generalized cross-entropy loss via (24) as the total loss, which can be expressed by

$$\mathcal{L}_{NS} = \mathcal{L}_{cons} + \mathcal{L}_{GCE} + \mathcal{L}_{kl}. \tag{27}$$

Additionally, we perform label correction based on the output of the teacher classifier via (26).

Weakly Supervised Learning: Weakly supervised learning aims to train models using coarser-grained, more readily available labels. Specifically, in this article, the term refers to situations where training samples correspond to a set of candidate labels due to label ambiguity. Given the ambiguious labeled data $D = \{(x_i, s_i)\}_{i=1}^{N_l}$, where s_i is the candidate label set. During training, label disambiguation is the key step, which refers to identifying the ground truth label from the candidate set. Before label disambiguation, it is necessary to initialize a label confidence vector $k_i = \{k_{i1}, k_{i2}, \ldots, k_{iM}\}$ for x_i as follows:

$$k_{ic} = \begin{cases} \dfrac{1}{|s_i|}, & if\ c \in s_i \\ 0, & otherwise. \end{cases} \tag{28}$$

Label disambiguation is to update the label confidence vector according to the predicted label probabilities p^s of student classifier, as expressed by the following formula:

$$k_{ic} = \begin{cases} \dfrac{p_{ic}^{s}}{\sum_{j\in s_i}\left(p_{ij}^{s}\right)}, & if\ j \in s_i \\ 0, & otherwise. \end{cases} \tag{29}$$

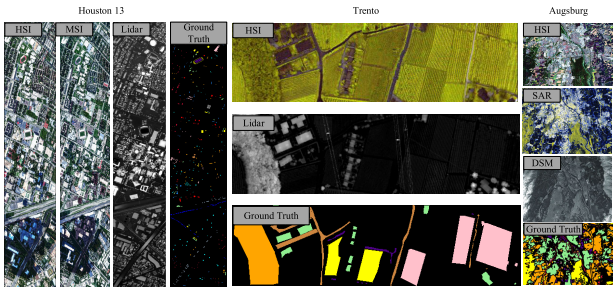


Fig. 4. Datasets used for experiments.

TABLE I
SUMMARY OF MULTIMODAL REMOTE SENSING DATASETS

Dataset	categories	Samples	modalities	Bands
Houston 13	15	15029	HSI/MSI/Lidar	144/8/1
Trento	6	30214	HSI/Lidar	63/1
Augsburg	7	78294	HSI/SAR/DSM	180/4/1
PC-PU	7	39355	HSI/HSI	102/102
PU-PC	7	39332	HSI/HSI	102/102
HS13-18	7	2530	HSI/HSI	48/48
HS18-13	7	53200	HSI/HSI	48/48

The cross-entropy loss is used for training, which is as follows:

$$\mathcal{L}_{ICE} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{M} k_{ic}\cdot\log\left(p_{ic}^{s}\right). \tag{30}$$

For the course supervision information, we initially obtain the label confidence vector via (28) and leverage the student's output to update it during the training process via (29). We leverage the sum of the feature contrastive loss $\mathcal{L}_{con}$, the consistency loss in (13), and the improved cross-entropy loss as the total loss, which can be expressed as follows:

$$\mathcal{L}_{WS} = \mathcal{L}_{kl} + \mathcal{L}_{con} + \mathcal{L}_{ICE}. \tag{31}$$

Unsupervised Learning: It does not utilize manually labeled data, allowing the model to directly discover the inherent structure within the data itself and cluster it. Since unsupervised learning lacks any labeled information, we directly utilize contrastive loss $\mathcal{L}_{cons}$ via (17) to optimize the learning process.

III. EXPERIMENTS

In this section, we will verify UniHyper on three multimodal datasets and compare with it some of the state-of-the-art land-cover classification algorithms.

A. Datasets

For the experiment of multimodal classification under diff-labeled scenes, we choose three public multimodal remote sensing image datasets, as shown in Table I and Fig. 4. The Houston 13 dataset comprises three different modalities: HSI with 144 spectral bands, MSI with eight channels, and LiDAR with one channel, and it has 349 × 1905 pixels with fifteen categories. The Trento dataset comprises HSI with 63 spectral bands and LiDAR with one channel, and it has 166 × 600 pixels with seven categories. The Augsburg dataset integrates HSI with 180 spectral bands, SAR with four channels, and

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TABLE II

EXPERIMENTAL RESULTS UNDER FULLY SUPERVISED AND SEMI-SUPERVISED SCENES ON HOUSTON 13 AND AUGSBURG DATASETS. "H" DENOTES THE HSI, "L" DENOTS LIDAR, "M" DENOTES THE MSI, "D" DENOTES THE DSM, AND "S" DENOTES THE SAR

Houston 13											
Metrics	Fully-Supervised							Semi-Supervised			
	MDL	EndNet	MTNet	DCMNet	H+L	H+M	**H+M+L**	PGS2Net	H+L	H+M	**H+M+L**
OA	89.60	88.52	92.66	95.11	96.12	95.28	**97.06**	95.36	96.76	96.09	**97.29**
κ	88.75	87.59	92.03	94.69	95.81	94.90	**96.82**	94.98	96.49	95.77	**97.07**
AA	90.83	89.95	93.51	95.74	96.04	95.49	**97.31**	95.02	96.67	96.29	**97.47**
Healthy grass(63)	83.1	81.58	83.1	80.72	96.47	95.46	**96.47**	95.54	97.60	98.31	**98.31**
Stressed grass(63)	81.58	83.65	99.44	98.78	98.57	99.49	**98.91**	99.66	97.79	97.43	**98.49**
Synthetic grass(35)	100	100	100	100.00	100.00	100.00	**100.00**	100.00	100.00	100.00	**100.00**
Trees(62)	99.72	93.09	100	99.53	98.73	99.15	**99.24**	99.66	97.95	96.43	**98.21**
Soil(62)	99.81	99.91	100	99.91	99.66	99.75	**99.75**	100.00	99.55	99.46	**99.55**
Water(16)	95.10	95.10	95.8	100.00	99.35	100.00	**100.00**	85.11	95.56	99.32	**97.95**
Residential(63)	90.02	82.65	92.44	85.45	98.01	97.93	**99.49**	99.50	94.75	95.36	**94.92**
Commercial(62)	87.94	81.29	96.68	89.36	92.72	87.56	**92.98**	95.94	95.27	92.95	**96.43**
Road(63)	81.59	88.29	86.69	97.73	96.13	90.50	**95.04**	97.14	94.14	91.66	**92.46**
Highway(61)	86.68	89.00	81.47	97.88	94.94	96.66	**97.68**	94.00	96.92	98.46	**99.46**
Railway(62)	89.37	83.78	93.26	97.15	96.08	92.33	**95.74**	90.55	95.41	93.62	**97.03**
Parking Lot 1(62)	85.69	90.39	85.40	99.52	89.51	92.41	**92.32**	80.03	97.57	94.14	**95.32**
Parking Lot 2(23)	83.16	82.46	88.42	89.82	81.84	82.51	**93.27**	88.12	88.89	87.23	**94.33**
Tennis Court(21)	100.00	100.00	100	100.00	100.00	99.75	**99.51**	100.00	100.00	100.00	**100.00**
Running Track(33)	98.73	98.10	100	100.00	98.56	98.88	**99.20**	100.00	98.65	100.00	**99.66**
Augsburg dataset											
Metrics	Fully-Supervised							Semi-Supervised			
	MDL-RS	EndNet	MTNet	DCMNet	H+S	H+D	**H+S+D**	PGS2Net	H+S	H+D	**H+S+D**
OA	79.11	78.25	83.51	87.68	90.07	91.29	**91.54**	91.69	90.56	92.01	**92.09**
κ	67.52	68.11	76.43	82.07	85.63	87.41	**87.76**	87.78	86.39	88.44	**88.57**
AA	46.74	52.29	57.32	58.39	67.74	71.05	**71.31**	59.62	68.69	69.64	**71.17**
Forest(146)	84.97	89.7	87.17	93.49	96.12	98.07	**96.90**	98.61	98.27	98.98	**98.58**
Residential-Area(264)	91.56	86.75	96.74	98.81	94.33	94.91	**95.20**	99.05	92.79	95.58	**95.09**
Industrial-Area(21)	8.17	24.26	66.92	16.84	51.75	52.41	**52.82**	44.86	62.17	58.94	**63.01**
Low-Plants(248)	78.07	76.77	78.18	89.86	95.26	96.31	**97.15**	96.25	96.17	97.25	**97.36**
Allotment(52)	26.00	34.42	45.70	45.51	77.06	86.81	**86.62**	37.86	77.28	83.01	**86.84**
Commercial-Area(7)	2.14	9.46	6.11	16.12	4.03	10.81	**14.22**	0.00	0.37	6.68	**5.95**
Water(23)	67.52	46.78	20.44	48.11	55.61	58.06	**56.27**	40.74	53.77	47.04	**51.35**

TABLE III

EXPERIMENTAL RESULTS UNDER FULLY SUPERVISED, SEMI-SUPERVISED, NOISY-SUPERVISED, AND WEAKLY SUPERVISED SCENES ON TRENTO DATASET. "H" DENOTES THE HSI AND "L" DENOTS LIDAR

Trento														
Metrics	Fully-Supervised					Semi-Supervised		Noisy-Supervised			Weakly-Supervised			
	MDL	EndNet	MTNet	DCMNet	**H+L**	PGS2Net	**H+L**	Dcrn	SPTNet	**H+L**	ParSe	PiCo	CrCD	**H+L**
OA	90.65	94.17	98.23	98.96	**99.03**	98.79	**99.14**	88.58	95.91	**97.92**	88.47	91.62	84.92	**98.00**
κ	86.28	92.22	97.62	98.61	**98.70**	98.39	**98.84**	84.94	94.54	**97.22**	84.64	89.20	79.85	**97.32**
AA	89.6	93.88	97.01	97.55	**98.28**	98.15	**98.63**	87.75	96.00	**97.27**	87.05	88.99	76.91	**97.09**
Apple trees(129)	88.58	88.19	96.7	99.38	**99.54**	99.72	**99.84**	96.26	98.68	**99.09**	90.91	89.55	78.10	**99.33**
Building(125)	95.86	98.49	99.28	94.39	**97.01**	96.69	**96.90**	79.26	89.10	**94.41**	83.30	79.41	64.40	**94.28**
Ground(105)	93.58	95.19	96.52	93.95	**97.86**	100	**99.26**	90.62	100.00	**98.76**	90.37	90.64	63.37	**98.12**
Woods(154)	99.22	99.30	99.91	100.00	**99.71**	99.90	**99.81**	96.63	100.00	**99.24**	89.54	94.88	86.99	**99.40**
Vineyard(184)	83.82	91.96	99.52	99.71	**99.81**	99.81	**99.69**	83.65	93.30	**98.53**	91.40	96.37	95.51	**98.89**
Roads(122)	76.51	90.14	90.14	98.61	**95.74**	92.76	**96.28**	80.10	94.94	**93.61**	76.80	83.09	73.07	**92.53**

digital surface model (DSM) data with one channel, and it contains 332×485 pixels with six categories.

B. Fully Supervised Classification

In the experiment of fully supervised classification, we select four representative multimodal methods for comparison: MDL [53], EndNet [54], MTNet [55], DCMNet [56]. The training sample number for each class on three datasets is listed in Tables II and III. As shown in Tables II and III, it is obvious that our method UniHyper is superior to the comparative methods. UniHyper obtained the highest overall accuracy (OA), AA, and κ values. Specifically, all comparative methods focus solely on designing distinct multimodal feature interaction and fusion approaches for HSI and LiDAR, which ignore the consistency and uniqueness among the features of different modalities. Under identical configurations, UniHyper outperforms the second-best algorithm (DCMNet) by over 3% on the Augsburg dataset. Furthermore, experimental results from both the Houston 13 and Augsburg datasets demonstrate that UniHyper consistently outperforms its two-modality variants when leveraging three modalities. All in all, compared to models that only consider just two modalities, our model has more advantages in handling multimodal data, which can process three-modality data simultaneously.

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TABLE IV

EXPERIMENTAL RESULTS UNDER NOISY-SUPERVISED AND WEAKLY SUPERVISED SCENES ON HOUSTON 13 AND AUGSBURG DATASETS. "H" DENOTES THE HSI, "L" DENOTES LIDAR, "M" DENOTES THE MSI, "D" DENOTES THE DSM, AND "S" DENOTES THE SAR

Houston 13											
Metrics	Noisy-Supervised					Weakly-Supervised					
	DCRN	SPTNET	H+L	H+M	**H+M+L**	ParSE	PiCO	CrCD	H+L	H+M	**H+M+L**
OA	71.36	87.95	88.89	86.38	**89.89**	81.03	85.41	94.05	95.93	95.18	**96.74**
κ	69.01	86.98	87.99	85.26	**89.06**	79.5	84.16	93.55	95.60	94.79	**96.47**
AA	73.01	89.57	89.83	87.39	**91.13**	81.03	85.57	94.22	96.05	95.43	**96.97**
Healthy grass	73.43	81.29	93.72	88.84	**94.96**	84.19	85.79	86.29	94.87	95.88	**95.96**
Stressed grass	84.35	99.34	98.22	99.17	**97.10**	70.39	75.92	92.11	99.41	99.33	**98.83**
Synthetic grass	85.37	100.00	99.69	100.00	**99.85**	92.16	94.27	96.08	100.00	100.00	**100.00**
Trees	63.86	99.83	93.77	88.95	**96.10**	77.33	85.79	90.02	98.65	97.72	**99.15**
Soil	85.65	91.18	98.59	99.08	**99.00**	91.1	94.92	99.75	99.49	99.58	**99.66**
Water	74.48	85.42	97.96	96.94	**99.66**	68.93	78.64	86.41	99.68	98.06	**99.35**
Residential	53.27	87.96	86.59	91.66	**90.68**	76.18	84.32	94.69	98.51	96.10	**98.67**
Commercial	64.70	66.53	68.35	59.57	**62.63**	65.06	69.04	84.69	92.05	86.97	**91.62**
Road	77.50	84.04	90.89	92.05	**86.34**	74.12	81.18	96.38	89.33	91.01	**92.35**
Highway	70.76	87.63	79.78	79.02	**93.34**	94.17	97.68	99.40	96.83	97.00	**96.14**
Railway	64.26	83.67	85.37	82.17	**86.54**	86.71	87.65	95.74	97.10	94.29	**96.51**
Parking Lot 1	53.10	77.45	80.84	69.87	**78.65**	79.61	86.95	97.53	91.13	89.93	**95.39**
Parking Lot 2	75.23	100.00	75.17	65.75	**82.99**	81.61	79.15	95.29	85.43	86.54	**92.38**
Tennis Court	82.05	100.00	99.75	100.00	**100.00**	92.63	98.03	100.00	99.75	100.00	**99.51**
Running Track	87.16	99.20	98.73	97.78	**99.05**	89.79	84.21	100.00	98.56	99.04	**99.04**
Augsburg											
Metrics	Noisy-Supervised					Weakly-Supervised					
	DCRN	SPTNET	H+S	H+D	**H+S+D**	ParSE	PiCO	CrCD	H+S	H+D	**H+S+D**
OA	80.45	84.15	71.23	78.87	**84.86**	76.81	74.24	74.52	88.28	89.45	**89.25**
κ	73.36	78.44	62.61	71.51	**79.15**	65.98	63.01	61.56	82.91	84.68	**84.44**
AA	75.40	84.40	71.58	75.72	**78.43**	48.48	47.99	37.70	57.30	62.24	**63.61**
Forest	87.38	93.28	94.29	94.13	**97.74**	75.44	72.41	60.74	93.87	97.52	**97.61**
Residential-Area	83.88	74.58	49.25	59.84	**74.06**	86.89	89.43	87.94	93.17	93.54	**92.93**
Industrial-Area	57.61	80.23	47.52	76.01	**72.33**	11.02	9.37	0.00	47.60	45.33	**42.61**
Low-Plants	77.26	91.48	87.99	94.64	**94.92**	83.34	76.25	86.11	94.99	95.72	**95.67**
Allotment	77.38	95.59	88.18	72.95	**72.34**	50.48	54.49	13.00	28.68	57.74	**64.44**
Commercial-Area	81.54	71.69	72.91	69.01	**66.26**	4.82	8.12	0.00	0.00	0.00	**1.40**
Water	62.76	83.92	60.89	63.44	**71.34**	27.34	25.88	16.12	42.80	45.85	**50.63**

C. Semi-Supervised Classification

In the experiment of semi-supervised classification, we added an equal number of unlabeled samples to the same number of precisely labeled samples used in the fully supervised experiments, and selected the latest semi-supervised algorithm PGS2Net [37] as the comparison algorithm. As shown in Tables II and III, it is evident that adding an appropriate amount of unlabeled samples to labeled samples of equal size can improve classification performance. Specifically, in the semi-supervised scene, UniHyper achieves up to 1% higher OA on the Houston 13 dataset compared to the fully supervised scene. Furthermore, UniHyper still holds a significant advantage over the latest semi-supervised algorithm (PGS2Net) on the three datasets. For example, UniHyper obtains 97.29%, 97.07%, and 97.47% in OA, κ, and AA on the Houston 13 dataset, which is 1.93%, 2.09%, and 2.45% higher than PGS2Net. The above findings indicate that UniHyper can more accurately utilize unlabeled samples to further improve the model performance.

D. Noisy-Supervised Classification

In the experiment of noisy-supervised classification, no ready specific sources of noisy labels are available for our analysis. To construct the mislabeled samples of these three hyperspectral datasets and to further simulate the problem of noisy labels, randomly selected samples from other classes are added to each class [39]. In this way, we randomly select 25 clean samples and 15 noisy samples for training on the Houston 13 dataset, and 75 clean samples and 25 noisy samples for training on the Trento and Augsburg datasets. Because of no corresponding multimodal method for noisy-supervised classification, we select two single-modal noisy-supervised methods, including DCRN [40] and SPTNet [42] for comparison. As shown in Tables III and IV, UniHyper still outperforms the existing methods under the noisy-supervised scene in most cases. It is noteworthy that on the large-scale Augsburg dataset, using only two modalities instead of three leads

TABLE V

CLASSIFICATION RESULTS UNDER WEAKLY SUPERVISED AND NOISY-SUPERVISED SETTINGS ON THE HOUSTON 13 DATASET. "R" IS THE PARTIAL-LABEL RATE AND "NL" IS THE NOISY-LABEL NUMBER OF EACH CLASS

Weakly Supervised					
R	ParSE	PiCo	CrCD	UniHyper-HSI	UniHyper
0.1	81.03	85.41	94.05	91.74	**96.74**
0.2	79.85	80.71	92.01	89.26	**96.07**
0.3	73.96	69.55	87.36	87.53	**95.65**
Noisy Supervised					
NL	DCRN	SPTNet	–	UniHyper-HSI	UniHyper
15	71.36	87.95	–	83.87	**89.89**
20	68.78	85.35	–	78.24	**89.52**
25	55.87	87.70	–	81.21	**89.23**

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TABLE VI

EXPERIMENTAL RESULTS UNDER UNSUPERVISED SCENE ON HOUSTON 13 AND TRENTO DATASETS

Datasets	Houston 13				Trento			
Method	OA	κ	PUR	NMI	OA	κ	PUR	NMI
MFLVC	42.08	37.58	44.71	39.55	77.88	71.37	80.10	61.39
GCFAgg	26.19	20.67	28.53	20.50	68.13	55.37	71.35	49.36
EMVCC	70.36	68.14	72.91	72.79	70.01	62.57	83.44	68.10
MDC	45.86	41.51	47.51	50.04	83.88	79.19	85.60	77.21
AMKSC	46.72	42.27	47.57	45.87	92.15	89.53	93.23	87.17
DGLAP	48.53	44.55	53.33	49.20	76.76	67.37	79.34	73.05
FPFC	39.43	34.81	44.19	42.36	93.34	91.14	93.34	87.68
UniHyper	**77.34**	**75.61**	**79.51**	**75.92**	**93.38**	**91.19**	**93.38**	**89.39**

TABLE VII

ABLATION EXPERIMENTS ON BACKBONE AND LOSS COMPONENTS. "S-CLASSIFER" MEANS THE SINGLE CLASSIFER WITHOUT TEACHER–STUDENT ARCHITECTURE

Methods	Components			Houston 13				
	Backbone	$\mathcal{L}_{cf}$	$\mathcal{L}_{uf}$	S1	S2	S3	S4	S5
UniHyper-A	CMT			87.30	87.42	76.57	85.59	65.21
UniHyper-B	FoCA			88.55	96.77	82.29	92.02	72.00
UniHyper-C	FoCA		✓	96.98	97.24	89.90	96.77	74.40
UniHyper-D	FoCA	✓		89.35	92.02	82.22	91.73	71.44
UniHyper-E	FoCA	✓	✓	**97.06**	**97.29**	**89.89**	**96.74**	**77.34**
Methods	Indentifier	$\mathcal{L}_{cf}$	$\mathcal{L}_{uf}$	S1	S2	S3	S4	S5
UniHyper-F	K-Means	✓	✓	–	–	–	–	72.51
UniHyper-G	S-classifer	✓	✓	97.04	97.21	88.01	96.72	–

TABLE VIII

COMPONENTWISE ABLATION STUDY OF FOCA-NET ON THE HOUSTON 13 AND TRENTO DATASETS

Components				Houston 13		Trento	
OF	FM	CA	MSCF	ACC	κ	ACC	κ
×	✓	✓	✓	96.28	95.98	98.22	97.62
✓	×	✓	✓	96.85	96.59	98.13	97.50
✓	✓	×	✓	96.12	95.81	98.41	97.88
✓	✓	✓	×	96.68	96.41	98.33	97.77
✓	✓	✓	✓	**97.06**	**96.82**	**99.03**	**98.70**

to a significant decline in classification performance. This clearly demonstrates the positive impact of multimodal data when annotation resources are limited. We further evaluate the modality-matched HSI-only UniHyper (UniHyper-HSI) under multiple partial-label and noisy-label settings. As shown in Table V, UniHyper maintains stable performance as the number of noisy labels increases, with ACCs of 89.89%, 89.52%, and 89.23% for 15, 20, and 25 noisy samples, respectively. In contrast, DCRN degrades substantially from 71.36% to 55.87%, while SPTNet also shows lower overall performance. The HSI-only variant remains competitive, indicating that the robustness is not solely due to additional modalities, while multimodal fusion further improves noise tolerance.

E. Weakly Supervised Classification

In the experiment of weakly supervised classification, following [44], we generate uniform partial labels by flipping each incorrect label with a probability R. We set R to 0.1 for the Houston 13 and Augsburg datasets, and 0.3 for the Trento dataset. The training size for each dataset is the same as the fully supervised scene. Because the existing methods for weakly supervised classification are mainly for single-modal, we select three single-modal algorithms for comparison: ParSE [57], PiCo [58], and CrCD [44]. As shown in Tables III and IV, it is obvious that UniHyper is evidently superior to ParSE, PiCo, and CrCD. Specifically, the classification performance of UniHyper on the Augsburg dataset is as high as 89.25%, 84.44%, and 63.61% in OA, κ, and AA, which is higher than the second-best algorithm (CrCD) by over 10%. We also add a modality-matched UniHyper-HSI and evaluate it under multiple partial-label settings. As shown in Table V, under weak supervision, the full UniHyper consistently achieves the best performance across all partial-label rates, with ACCs of 96.74%, 96.07%, and 95.65% at rates of 0.1, 0.2, and 0.3, respectively. Notably, its performance decreases by only 1.09% as label ambiguity increases, whereas CrCD drops by 6.69%. The HSI-only variant also remains competitive and slightly surpasses CrCD at the highest ambiguity level (87.53% versus 87.36%), indicating that the proposed learning strategy itself provides robustness to ambiguous supervision, while multimodal information further improves performance.

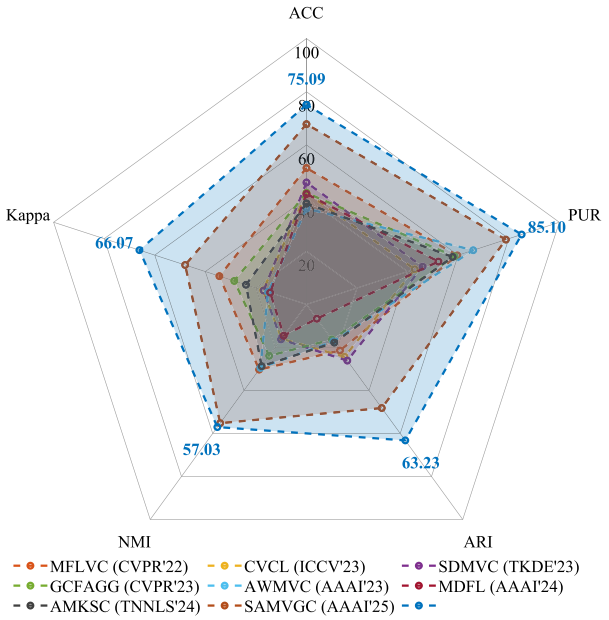


Fig. 5. Comparison on the Augsburg dataset under unsupervised scene.

F. Unsupervised Classification

In the experiment of unsupervised classification, we choose seven representative clustering methods: MFLVC [59], GCFAgg [60], EMVCC [61], MDC [62], AMKSC [63], DGLAP [46], and FPFC [47] for comparison on the Houston 13 and Trento datasets. For large-scale dataset augsburg, we additionally selecte CVCL [64], SDMVC [65], AWMVC [66], MDFL [67], and SAMVGC [68] for further comparison. In addition to the commonly used classification metrics OA and κ, we also employed the unsupervised metrics Purity (PUR) and NMI to evaluate the effectiveness of the models. As shown in Table VI and Fig. 5, compared with the existing clustering algorithms, UniHyper consistently achieves the best clustering results on most datasets. For example, on the Houston 13 dataset, UniHyper outperforms the second-best method by 6.98%, 7.47%, 6.6%, and 3.13% in ACC, κ, PUR,

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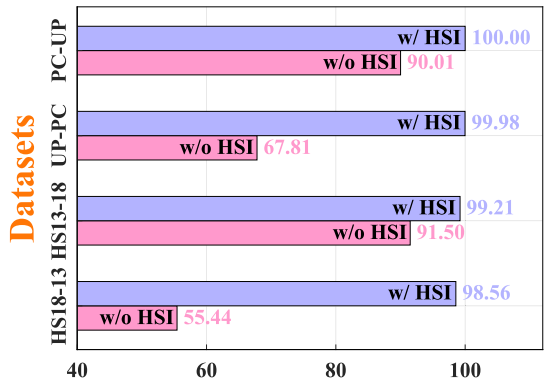


Fig. 6. Clustering accuracy (ACC) on four HSI-HSI datasets. The chart compares performance without HSI (single-modal, labeled w/o HSI, in pink) and with HSI (multimodal, labeled w/HSI, in light purple).

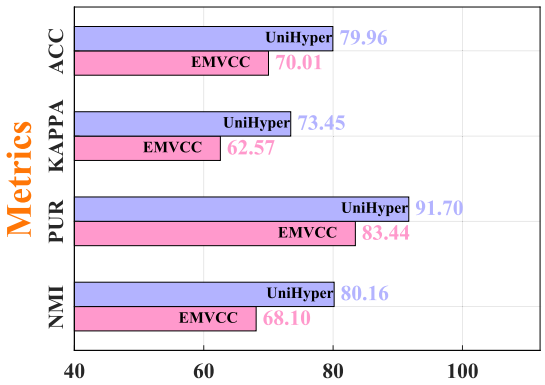


Fig. 7. Clustering accuracy on four Houston 13 datasets. The chart compares performance for multiview classification under the unsupervised scene.

and NMI, respectively. We can also see the obvious advantage of UniHyper on the Augsburg dataset. The above data fully validates the effectiveness of UniHyper under unsupervised scene.

G. Extended Classification Scene

Since feature representation quality is a key factor affecting performance across different supervision regimes, UniHyper aims to provide a unified multimodal framework adaptable to diverse supervision scenarios. To further verify the generalization capability of the learned representations, we extend the evaluation to cross-scene and multiview clustering tasks under the unsupervised setting.

For the cross-sensor, as shown in Fig. 6, we pair samples according to their common land-cover categories and evaluate representation learning across different hyperspectral sensors. For example, the Pavia University (PU, 610×315 pixels) and Pavia Center (PC, 1096×715 pixels) with seven shared classes after removing noninformative samples. Similarly, Houston 13 (210×954 pixels, 48 bands) and Houston 18 (210×954 pixels, 48 bands) are paired with seven shared classes. There are four cross-hyperspectral sensor datasets: UP-PC, PC-UP, HS13-18, and HS18-13, shown in Table I. As shown in Fig. 4, it is evident that incorporating the HSI from different sensors leads to a dramatic improvement in OA across all cross-sensor datasets. Specifically, classification accuracy increases from 90.01% (w/o HSI) to 100% under unsupervised scenes on PC-UP. Similarly, for HS13-18, from 91.50% (w/o HSI) to 99.21%. These experimental results clearly demonstrate UniHyper's universality for cross-sensor scenarios.

For the multiview scenario, as shown in Fig. 7, we construct spectral view and spatial view of HSI provided by the Houston 13 dataset with a single modality, and select the multiview clustering method EMVCC as a comparative modal. As shown in Fig. 5, it is evident that UniHyper can extract valuable information from different perspectives and obtains better performance when processing unimodal data. Specifically, compared with EMVCC, UniHyper improves the classification performance from 70.11% to 79.96% in terms of OA, and from 62.57% to 73.45% in terms of κ. This further demonstrates the applicability of feature decoupling theory to information from different perspectives within the same modality.

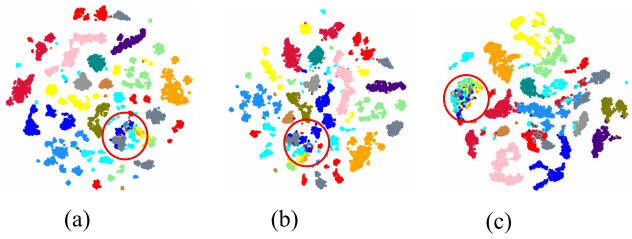


Fig. 8. t-SNE visualization of full, unique, and common representation spaces on Houston 13. (a) Full embedding ACC: 77.34. (b) Unique subspace ACC: 45.15. (c) Common subspace ACC: 61.19.

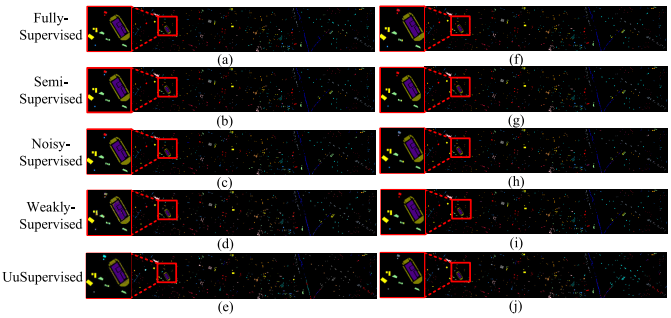


Fig. 9. Classification maps on Houston 13 dataset under diff-labeled scenarios. (a) DCMNet. (b) PGS2Net. (c) SPTNet. (d) ParSE. (e) EMVCC. (f) UniHyper. (g) UniHyper. (h) UniHyper. (i) UniHyper. (j) UniHyper.

H. Visualization Analysis

Fig. 8 illustrates the learned full embedding, common subspace, and unique subspace. The common subspace exhibits compact and well-separated clusters, demonstrating effective alignment of cross-modal shared semantics. In contrast, the unique subspace preserves modality-unique diversity with a more dispersed distribution, indicating successful feature decoupling without semantic collapse. By integrating shared and modality-unique representations, the full embedding achieves more discriminative structures and superior clustering performance. These results verify that UniHyper effectively disentangles common and unique information in a complementary manner.

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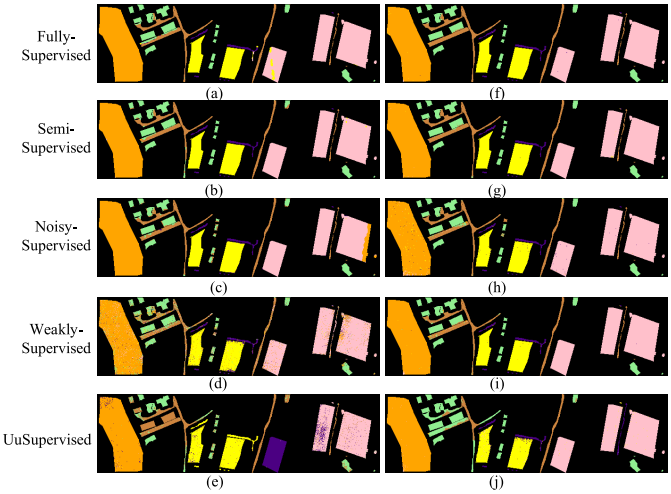


Fig. 10. Classification maps on trento dataset under diff-labeled scenarios. (a) DCMNet. (b) PGS2Net. (c) SPTNet. (d) ParSE. (e) EMVCC. (f) UniHyper. (g) UniHyper. (h) UniHyper. (i) UniHyper. (j) UniHyper.

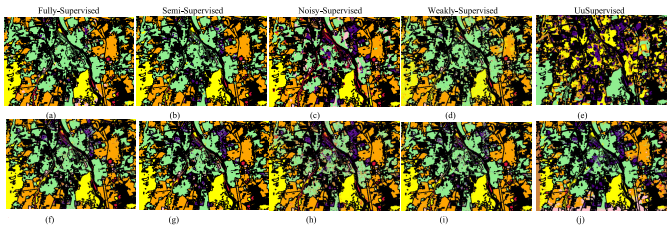


Fig. 11. Classification maps on the Augsburg dataset under diff-labeled scenarios. (a) DCMNet. (b) PGS2Net. (c) SPTNet. (d) ParSE. (e) AMKSC. (f) UniHyper. (g) UniHyper. (h) UniHyper. (i) UniHyper. (j) UniHyper.

Figs. 9–11 show the visualized classification diagrams of different methods under diff-labeled scenarios on the Houston 13, Terento, and Augsburg datasets, respectively. As can be seen from the figure, compared with the baseline methods under diff-labeled scenarios, the land-cover area of the same category obtained by our proposed UniHyer is more uniform and has the least misclassified pixels. It is particularly evident in areas where mixed ground objects have similar spectral characteristics. Our method introduces an advanced multimodal feature decoupling method to extract valuable and discriminative features of different land covers. As a result, UniHyper can better handle complex structures and retain more details for land-cover identification.

I. Ablation Analysis

The ablation experiment in Table VII reveals the contribution and interaction mechanism of each module. First, by comparing UniHyerA and UniHyerB on diff-labeled scenarios (S1 is fully supervised, S2 is semi-supervised, S3 is noisy-supervised, S4 is weakly supervised, and S5 is unsupervised), we observe that using FoCA as the backbone network improves ACC by 6.79% compared to CMT, which confirms the effectiveness of FoCA. Second, by comparing UniHyerB and UniHyerC, the ACC can be further improved via introducing $\mathcal{L}_{uf}$, which denotes that we can improve the classification performance via constraining the unique features, reducing feature redundancy, and enhancing modal specificity. It is worth noting that in UniHyerD, when $\mathcal{L}_{cf}$ is used alone, it results in a decrease in the ACC by 1%, which suggests that overconstraining the common features may compress intraclass diversity, highlighting the need for optimization in conjunction with $\mathcal{L}_{uf}$. Finally, by comparing UniHyerE with UniHyerF and UniHyper-G, it is evident that the student–teacher classifier and dynamic prototype clustering are more effective for land-cover classification. UniHyerE demonstrates the synergistic interaction among the feature encoder, feature decoupling strategy, and the identifier proposed in this article.

To further clarify the contribution of each component in FoCA-Net, we conducted componentwise ablations of the OF, Fourier masking (FM), channel attention (CA), and MSCF on both Houston 13 and Trento under fully supervised sense. As shown in Table VIII, removing any individual component consistently degrades performance. On Houston 13, the ACC decreases from 97.06% to 96.28%, 96.85%, 96.12%, and 96.68%. Similar trends are observed on Trento, where the full model achieves 99.03% ACC, compared with 98.22%, 98.13%, 98.41%, and 98.33% after removing each component. These results demonstrate that all four components contribute positively to FoCA-Net and that their effects are consistent across datasets.

TABLE IX
PERFORMANCE STABILITY OVER FIVE INDEPENDENT RUNS ON THE HOUSTON 13 DATASET

Metrics	Fully Supervised		Semi-Supervised	
	DCMENT	UniHyper	PGS2Net	UniHyper
OA	92.36 ± 2.22	$\mathbf{97.10 \pm 0.40}$	95.96 ± 0.85	$\mathbf{97.49 \pm 0.24}$
κ	91.75 ± 2.38	$\mathbf{96.87 \pm 0.43}$	95.63 ± 0.93	$\mathbf{97.26 \pm 0.26}$
Metrics	Weakly Supervised		Noisy Supervised	
	CrCD	UniHyper	SPTNet	UniHyper
OA	93.18 ± 1.24	$\mathbf{96.22 \pm 0.49}$	88.77 ± 1.15	$\mathbf{89.53 \pm 0.77}$
κ	92.62 ± 1.32	$\mathbf{95.91 \pm 0.53}$	87.86 ± 1.24	$\mathbf{88.69 \pm 0.83}$
Metrics	Unsupervised			
	EMVCC		UniHyper	
OA	71.41 ± 1.48		$\mathbf{76.58 \pm 1.12}$	
κ	69.27 ± 1.60		$\mathbf{74.78 \pm 1.21}$	

J. Stability Analysis

To further address the concern regarding experimental stability, we additionally conducted five independent runs on Houston 13 under all five supervision settings and report mean $\pm$ standard deviation. As shown in Table IX, UniHyper achieves 97.10 $\pm$ 0.40, 97.49 $\pm$ 0.24, 96.22 $\pm$ 0.49, 89.53 $\pm$ 0.77, and 76.58 $\pm$ 1.12 OA under fully supervised, semi-supervised, weakly supervised, noisy-supervised, and unsupervised settings, respectively. In all cases, UniHyper consistently outperforms the corresponding representative baseline while exhibiting relatively small variance. These results demonstrate that the reported improvements are stable across repeated runs rather than being caused by a particular random split or initialization.

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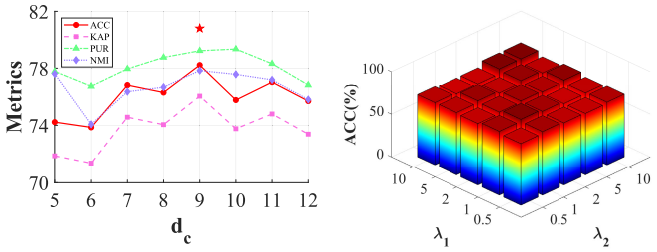


Fig. 12. (Left) Impact of different d_c values on clustering performance on the Houston 13 dataset. (Right) The effect of varying λ_1 and λ_2 parameters on ACC (%) on the Houston 13 dataset.

TABLE X
SENSITIVITY ANALYSIS OF THE SUBSAMPLING RATIO ρ ON THE HOUSTON 13 DATASET

ρ	ACC	KC	NMI	PUR
0.05	76.06	74.24	75.06	78.20
0.10	**77.34**	**75.61**	**75.92**	**79.51**
0.20	75.69	73.80	74.81	76.80
0.30	73.04	70.93	73.70	73.53

K. Parameter Analysis

We study the impact of several key parameters on the Houston 13 datasets under unsupervised scenarios, including the dimension d_c of the common features and the loss weights λ_1 and λ_2 of the feature decoupling loss.

Dimension d_c of Common Features: The plotted curves in Fig. 12 illustrate how clustering performance metrics vary with the common feature dimension d_c on the Houston 13 dataset. As d_c increases from 5 to 12, the ACC shows an upward trend, peaking around $d_c = 9$, before declining. Similar trends are observed for κ and NMI, while PUR exhibits a moderate improvement, with a clear peak also near $d_c = 9$. The maximum ACC value is highlighted by a pentagon marker, emphasizing the optimal performance achieved at this dimension. Overall, the graph demonstrates that setting d_c to approximately 9 provides the best balance between retaining cross-modal consistency and preserving modality-unique details, leading to superior clustering outcomes.

Parameters λ_1 and λ_2: The parameters $\lambda_1, \lambda_2 \in \{0.5, 1, 2, 5, 10\}$ determine the weight of each part of the loss ($\mathcal{L}_{cf}$ and $\mathcal{L}_{uf}$) in the overall loss. Through analysis on the Houston 13 dataset, we determine that when $\lambda_1 = 10$ and $\lambda_2 = 5$, the clustering accuracy reaches 77.34%.

Subsampling Ratio ρ: We further evaluate the sensitivity to the subsampling ratio ρ, which is listed in Table X. As ρ varies from 0.05 to 0.30, the ACC values are 76.06%, 77.34%, 75.69%, and 73.04%, respectively, with $\rho = 0.1$ achieving the best overall performance (77.34% ACC, 75.61% Kappa, 75.92% NMI, and 79.51% PUR). These results show that the clustering performance remains relatively stable over a reasonable range of subsampling ratios, while $\rho = 0.1$ provides a good balance between efficiency and performance.

Parameters η: Table XI presents the sensitivity analysis of the FoCA masking ratio on the Houston 13 dataset. The model achieves the best overall performance at a masking ratio of 20%, yielding the highest ACC, κ, ARI, and PUR scores. As the masking ratio further increases, the performance gradually deteriorates, suggesting that excessive masking may suppress informative low-frequency components and weaken discriminative representation learning. These observations demonstrate that an appropriate masking ratio effectively balances feature preservation and frequency-domain filtering, thereby enhancing the robustness of UniHyer.

TABLE XI
MASKING RATIO η SENSITIVITY ON HOUSTON 13

Mask ratio	ACC	κ	NMI	ARI	PUR
0%	75.59	73.77	76.66	66.01	78.73
20%	77.34	75.61	75.92	67.19	79.51
40%	74.23	72.26	75.14	63.13	75.55
60%	72.64	70.59	73.28	60.69	74.78
80%	73.34	71.30	73.56	60.21	74.72
90%	75.35	73.36	73.94	61.45	77.78

TABLE XII
PERFORMANCE AND COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT BACKBONES ON THE TRENTO DATASET

Method	Params (M)	FLOPs (G)	NMI	PUR	κ	OA
RGSA	13.00	9.22	68.31	75.48	65.56	75.11
TKSA	14.32	11.20	73.23	83.87	76.63	82.40
RMT	**10.15**	13.77	87.60	91.71	88.24	91.20
CMT	44.03	24.32	72.48	84.61	78.95	84.26
OrthoNets	51.51	34.05	86.14	92.23	89.15	91.84
UniHyper	14.31	**7.44**	**89.39**	**93.38**	**91.19**	**93.38**

L. Classification Results and Running Time

As shown in Table XII, we compare FoCA-Net with other state-of-the-art backbone networks, including RGSA [69], TKSA [70], RMT [71], CMT [48], and OrthoNets [49] on the Trento dataset. The results demonstrate that FoCA-Net achieves competitive clustering performance while significantly reducing the number of parameters and FLOPs, highlighting its lightweight and efficient design.

IV. CONCLUSION

In this article, we propose a unified multimodal classification framework for HSI, termed UniHyper, adaptive to land-cover classification under diff-labeled scenes. First, we design a FoCA-Net to achieve feature extraction among diverse modalities. Then, we introduce a Fourier-orthogonal feature decoupling strategy to alleviate the specificity erosion problem via decoupling modality-invariant and modality-unique features, and obtain effectively discriminative features for classification. Furthermore, to obtain satisfactory classification under diff-labeled scenarios, we introduce the teacher–student classifier for land-cover classification with some supervision information, and the dynamic prototype clustering strategy for unsupervised land-cover classification. The extensive experiments have demonstrated the model's effectiveness. The proposed UniHyper framework not only enhances feature consistency across modalities but also preserves modality-unique characteristics to extract discriminative features conducive to classification. This framework achieves classification universality across different labeling scenarios, thereby enhancing the applicability of classification models.

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In subsequent work, we will further explore more effective feature decoupling methods and extend the model's universality to different remote sensing tasks.

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